

Validity Analysis: GERMANQUAD_GERMANDPR

Target context: German-Speaking E-Commerce Shoppers (DACH)

Overall risk: **HIGH**

Deployment Context

Use case: A German online retailer deploys a customer-facing chatbot on its e-commerce website and mobile app to handle tier-1 support inquiries in the customer service and retail operations domain. Customers type informal, often terse questions (for example, "Wo ist meine Bestellung?" or "Kann ich das umtauschen ohne Rechnung?") about order status, return eligibility, and refund timelines. The chatbot extracts a direct answer from relevant passages from the retailer's internal FAQ and terms-of-service documents, which is shown to the customer without human review unless the query is escalated. The system's output directly determines whether a customer proceeds with a return, waits for a shipment, or gets routed to a live agent, so extraction accuracy affects both customer satisfaction and the volume of tickets escalated to human staff during peak periods.

Target population: Everyday online shoppers across Germany, Austria, and German-speaking Switzerland who contact e-commerce customer support via web chat or mobile app, spanning a broad age and income range but skewing toward mobile-first users in their 20s to 40s who type quickly and informally on smartphones.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ■	2	Significant gaps	high
Input Content ■	1	Serious concern	high
Input Form	2	Significant gaps	high
Output Ontology ■	2	Significant gaps	high
Output Content ✓	3	Moderate gaps	medium
Output Form ✓	3	Moderate gaps	high
Average	2.2		

■ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

GermanQuAD/GermanDPR is a well-engineered, transparently documented Standard-German QA/retrieval benchmark whose construct is Wikipedia-based encyclopedic extractive QA. For a DACH Tier-1 e-commerce customer support chatbot handling procedural queries with no human review, five of six validity dimensions show substantial gaps: taxonomy omits procedural, unanswerable, and escalation categories (IO); content is Wikipedia rather than FAQ/ToS with no coverage of DE/AT/CH legal variation (IC); questions are formal reformulated Standard German rather than terse mobile SMS-register (IF); output space has no abstain/escalate class (OO); metrics do not measure abstain or task completion (OF). The strongest dimension is output content, where a rigorous three-way annotation process with quality controls provides methodological credibility, though the annotator pool (NLP students plus one expert) is not representative of DACH shoppers. Deployment carries additional legal exposure under GDPR Art. 22 and the EU AI Act that the benchmark cannot address.

Practical Guidance

What This Benchmark Measures

The benchmark measures a model's ability to extract fact-seeking answer spans from formal Standard-German encyclopedic prose and to retrieve relevant German Wikipedia passages against a 2.8M-passage index. Its strongest dimension for this deployment is output content methodology — the three-way test annotation with expert QA [Q41–Q47] and Top-1-Accuracy reporting [Q62] provide a credible baseline for how well a model handles well-formed German extractive QA in general.

Construct Depth

Depth of evidence is high for the narrow construct (Wikipedia-domain extractive QA in formal Standard German) and low for the deployment construct (retail-FAQ procedural QA with abstain/escalate behaviour under DACH-jurisdictional constraints). The benchmark's own authors flag the metric-sensitivity issue for semantically-correct answers [Q83–Q85], and the taxonomy contains no unanswerable class [WEB-10]. Passing this benchmark provides evidence about baseline German extractive competence but no evidence about (a) procedural query handling, (b) safe abstention, (c) legal-content correctness across DE/AT/CH, or (d) informal-register robustness.

What Else You Need

Required supplements: (1) a retailer-specific FAQ/ToS evaluation set with procedural queries and gold answers reviewed by CS agents (addresses IC, OC gaps); (2) an unanswerable-question extension following the SQuAD v2.0 paradigm [WEB-10] with retailer-realistic no-answer cases (addresses IO, OO gaps); (3) an informal-register perturbation set with mobile-typed, typo-heavy, Umlaut-substituted, and Austrian/Swiss-vocabulary variants (addresses IF gap); (4) an escalation-accuracy and task-completion metric, potentially using MOVE-style embedding/LLM-judge scoring [WEB-12] (addresses OF gap); (5) a jurisdiction-aware content audit ensuring Swiss users are not given EU-only return-rights answers [WEB-16] and that § 356a BGB Widerrufsbutton content is current [WEB-14].

ID	Evidence
Q41	"We construct a three-way annotated test set including 2204 questions." (p.3)
Q42	"Student annotators formulate all questions and mark the first answer." (p.3)
Q43	"The second answer comes from our expert annotator who also removes questions that are not formulated clearly." (p.3)
Q44	"For the third annotation, we use the crowd annotation provider Crowd Guru." (p.3)
Q45	"Crowd Guru's annotations are checked for positional overlap with existing answers." (p.3)
Q46	"If there is no overlap, we manually check for irregularities and as a result, remove 76 wrong answers." (p.3)
Q47	"The final dataset consists of 2204 questions with $2204 \cdot 3 - 76 = 6536$ answers." (p.3)
Q62	"Figure 3 shows the impact of the number of training samples on the percentage of exact matches, F1-score, and Top-1-accuracy (any overlap of prediction and ground truth) on the GermanQuAD test set. For all three metrics, there are no improvements after using more than 80 percent of the training samples." (p.4)
Q83	"A promising path for future work is to research new evaluation metrics for QA." (p.7)

Q84	“Especially for longer answers in German, we find that current metrics penalise too much if a predicted answer contains some extra words although being semantically similar to the ground-truth answer.” (p.7)
Q85	“Future metrics could take into account the semantic similarity of predicted and ground-truth answers.” (p.7)
WEB-10	SQuAD 2.0 defines the unanswerable-question output class the benchmark lacks (https://arxiv.org/abs/1806.03822)
WEB-12	MÖVE uses embedding- and LLM-based metrics for German FAQ-style QA — an alternative metric approach relevant to the deployment (https://arxiv.org/abs/2606.13111)
WEB-14	§ 356a BGB requires a digital Widerrufsbutton with two-step confirmation for German online B2C (https://www.freshfields.com/en/our-thinking/blogs/risk-and-compliance/pitfalls-for-e-commerce-how-the-new-eu-withdrawal-button-widerrufsbutton-wi-102ms91)
WEB-16	Switzerland is not covered by Directive 2011/83/EU; return rights follow Swiss OR and retailer policy (https://www.iubenda.com/en/blog/understanding-the-right-of-withdrawal-in-the-eu-a-guide-for-online-businesses/)

Input Ontology (IO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: GERMANQUAD_GERMANDPR | Context: German-Speaking E-Commerce Shoppers (DACH)

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's taxonomy is scoped to extractive span-selection QA over encyclopedic Wikipedia content [Q4, Q87] and passage retrieval [Q5], with annotators explicitly encouraged to produce complex, multi-entity fact-seeking questions [Q22, Q87]. The DACH e-commerce deployment requires categories the benchmark does not cover: procedural/conditional queries (return eligibility, refund timelines), status lookups, and — critically — a decision category for abstain/escalate when the FAQ/ToS does not contain an answer. The unanswerable-question category central to SQuAD v2.0 [WEB-10] has no German counterpart in this benchmark, and the deployment's no-human-review setup makes this omission operationally severe. Construct underrepresentation is therefore significant.

ID	Evidence
Q4	"An extractive QA model trained on GermanQuAD significantly outperforms multilingual models and also shows that machine-translated training data cannot fully substitute hand-annotated training data in the target language." (p.1)
Q5	"We demonstrate the wide range of applications of GermanQuAD by adapting it to GermanDPR, a training dataset for dense passage retrieval (DPR), and train and evaluate one of the first non-English DPR models." (p.1)
Q22	"We encourage creation of complex questions that cannot be answered with a single entity." (p.2)
Q87	"A good question is a fact-seeking question that can be answered with an entity (person, organisation, location, etc.) or explanation. A bad question is ambiguous, incomprehensible, dependent on clear false presuppositions, opinion seeking, or not clearly a request for factual information." (p.8)
WEB-10	SQuAD 2.0 defines the unanswerable-question output class the benchmark lacks (https://arxiv.org/abs/1806.03822)

Strengths

- Covers two complementary task types (extractive QA and dense retrieval) that together approximate an open-domain FAQ pipeline architecturally relevant to the deployment [Q4, Q5]
- Question types are documented by frequency with worked examples [Q55], giving the deployer a clear picture of what taxonomy is (and is not) covered

ID	Evidence
Q4	"An extractive QA model trained on GermanQuAD significantly outperforms multilingual models and also shows that machine-translated training data cannot fully substitute hand-annotated training data in the target language." (p.1)
Q5	"We demonstrate the wide range of applications of GermanQuAD by adapting it to GermanDPR, a training dataset for dense passage retrieval (DPR), and train and evaluate one of the first non-English DPR models." (p.1)
Q55	"Table 2 lists question types with examples sorted by frequency in the test set of GermanQuAD." (p.3)

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories

IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Deployment requires: (a) procedural queries about return eligibility, order status, refund timelines, exchange without receipt, delivery exceptions; (b) escalation-to-agent routing decisions; (c) abstain behaviour when the source FAQ/ToS does not contain the answer. Grounded in deployment_context and query_intent_distribution in the region YAML.

IO-2: Yes. The benchmark's taxonomy consists of fact-seeking encyclopedic questions [Q87] and omits procedural/conditional customer-support queries. It also contains no unanswerable questions, so abstain/escalate is not a category in the ontology (contrast with SQuAD v2.0 [WEB-10]).

IO-3: The benchmark's complex-multi-entity encyclopedic question category [Q22] is not directly harmful but is largely irrelevant to the deployment's procedural query mix — evaluation weight is spent on a distribution that does not resemble deployment traffic.

IO-4: Documented gaps: (1) no procedural/conditional query category; (2) no unanswerable/abstain category; (3) no escalation-routing decision category. These are content-validity gaps for the specified deployment.

ID	Evidence
Q4	"An extractive QA model trained on GermanQuAD significantly outperforms multilingual models and also shows that machine-translated training data cannot fully substitute hand-annotated training data in the target language." (p.1)
Q22	"We encourage creation of complex questions that cannot be answered with a single entity." (p.2)
Q87	"A good question is a fact-seeking question that can be answered with an entity (person, organisation, location, etc.) or explanation. A bad question is ambiguous, incomprehensible, dependent on clear false presuppositions, opinion seeking, or not clearly a request for factual information." (p.8)
WEB-10	SQuAD 2.0 defines the unanswerable-question output class the benchmark lacks (https://arxiv.org/abs/1806.03822)
WEB-12	MÖVE uses embedding- and LLM-based metrics for German FAQ-style QA — an alternative metric approach relevant to the deployment (https://arxiv.org/abs/2606.13111)

Information Gaps

- No empirical breakdown of deployment query-type distribution to quantify how much benchmark taxonomy overlaps with actual traffic

Requires Expert Verification

- Whether any subset of GermanQuAD question types (e.g., 'how' or 'when' subclasses in Table 2) can serve as a proxy for procedural queries

Remediation

Gap 1: No procedural, unanswerable, or escalation-routing query categories in the benchmark taxonomy

Recommendation: Construct a supplementary retailer-realistic evaluation set covering the deployment's query-intent distribution (order status, return eligibility, refund timelines, exchange without receipt, delivery exceptions) plus an unanswerable-question extension modeled on SQuAD v2.0 [WEB-10] and MÖVE FAQ curation [WEB-12], including a distinct 'requires escalation' category.

ID	Evidence
WEB-10	SQuAD 2.0 defines the unanswerable-question output class the benchmark lacks (https://arxiv.org/abs/1806.03822)
WEB-12	MÖVE uses embedding- and LLM-based metrics for German FAQ-style QA — an alternative metric approach relevant to the deployment (https://arxiv.org/abs/2606.13111)

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: GERMANQUAD_GERMANDPR | Context: German-Speaking E-Commerce Shoppers (DACH)

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

All passages are sourced from German Wikipedia [Q26], with hard negatives from a full German Wikipedia dump [Q13, Q58]. The authors acknowledge that Wikipedia bias is difficult to overcome [Q37] and that the dataset focuses on factual rather than opinion questions [Q39]. The deployment context requires reading of internal FAQ and terms-of-service documents — procedural, conditional, list-heavy legal-register prose — for which Wikipedia is a poor proxy (domain_mismatch_severity: Major). The only out-of-domain generalization test uses a financial compliance dataset (SFCR) [Q74, Q75], not retail. Regional legal content (BGB §356a Widerrufsbutton [WEB-14], Swiss OR distinctions [WEB-16]) is entirely absent, meaning the benchmark cannot detect whether a model correctly handles jurisdictionally-varying return rights across DE/AT/CH.

ID	Evidence
Q13	"By adding hard negatives selected from the whole German Wikipedia to GermanQuAD, we create a training dataset in DPR format, which we call GermanDPR." (p.1)
Q26	"The basis of GermanQuAD is the German counterpart of English Wikipedia articles' used in SQuAD." (p.2)
Q37	"The bias inherent in Wikipedia texts is difficult to overcome when labelling extractive QA on these texts." (p.3)
Q39	"We hypothesize questions seeking factual information instead of opinions make the dataset less prone to bias." (p.3)
Q58	"We take GermanQuAD as a starting point and add hard negatives from a dump of the full German Wikipedia. Following the approach by Karpukhin et al. (2020), we use BM25 to find passages in Wikipedia that are most similar to the question but do not contain the answer string." (p.4)
Q74	"In order to test how well models trained on GermanQuAD generalize to other data, we evaluate them on the German part of MLQA (Lewis et al., 2020a), XQuAD (Artetxe et al., 2020), and a private dataset on financial compliance reports called SFCR." (p.6)
Q75	"SFCR is annotated SQuAD-style and consists of 2386 QA pairs." (p.6)
WEB-14	§ 356a BGB requires a digital Widerrufsbutton with two-step confirmation for German online B2C (https://www.freshfields.com/en/our-thinking/blogs/risk-and-compliance/pitfalls-for-e-commerce-how-the-new-eu-withdrawal-button-widerrufsbutton-wi-102ms91)
WEB-16	Switzerland is not covered by Directive 2011/83/EU; return rights follow Swiss OR and retailer policy (https://www.iubenda.com/en/blog/understanding-the-right-of-withdrawal-in-the-eu-a-guide-for-online-businesses/)

Strengths

- Train/test overlap is low (12.3%) relative to other QA datasets [Q50, Q51], indicating memorization is not driving reported performance
- Passages are authentic German Wikipedia prose rather than machine-translated content, so language quality is native [Q8, Q26]
- Authors are transparent about the Wikipedia-source bias limitation [Q37, Q39]

ID	Evidence
Q8	"Translations can also be a source of errors (some translated answers contain non-negligible errors) and can have a negative impact on the variety of language use that might limit model performance." (p.1)
Q26	"The basis of GermanQuAD is the German counterpart of English Wikipedia articles' used in SQuAD." (p.2)

Q37	"The bias inherent in Wikipedia texts is difficult to overcome when labelling extractive QA on these texts." (p.3)
Q39	"We hypothesize questions seeking factual information instead of opinions make the dataset less prone to bias." (p.3)
Q50	"For 12.3% of the questions in the test set, one of their answers also occurs in the training set." (p.3)
Q51	"This number is low compared to 58% to 72% reported by Lewis et al. (2020b) for other QA datasets." (p.3)

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Deployment queries require region-specific knowledge of EU/German/Austrian consumer law (14-day Widerrufsrecht, § 356a BGB Widerrufsbutton [\[WEB-13, WEB-14\]](#)) and Swiss OR-based return regimes [\[WEB-16\]](#), plus retailer-specific FAQ/ToS content. None of this is in the benchmark's Wikipedia corpus.

IC-2: Culturally-sensitive alignment is not directly evaluated; the corpus is encyclopedic factual content [\[Q39\]](#). Regional legal/regulatory content critical to answer correctness for return-eligibility queries is absent.

IC-3: The benchmark does not contain Western-specific content that misleads, but its Wikipedia domain does not transfer to procedural retail FAQ content — the mismatch is a domain gap rather than a Western-values gap.

IC-4: INSUFFICIENT DOCUMENTATION — no regional annotator recruitment for retail-domain content is described; the annotator pool is students plus one expert [\[Q18\]](#), none identified as shoppers or CS agents.

IC-5: Documented content-validity issues: (1) domain mismatch (Wikipedia vs. FAQ/ToS); (2) absence of Austrian and Swiss legal-context content [\[WEB-16\]](#); (3) absence of § 356a BGB Widerrufsbutton material [\[WEB-14\]](#); (4) private SFCR OOD test is financial compliance, not retail [\[Q74, Q75\]](#).

ID	Evidence
Q18	"We collaborate with students in the area of computational linguistics, computer science, and chemistry and one expert annotator who created more than 20k QA annotations." (p.2)
Q26	"The basis of GermanQuAD is the German counterpart of English Wikipedia articles' used in SQuAD." (p.2)
Q37	"The bias inherent in Wikipedia texts is difficult to overcome when labelling extractive QA on these texts." (p.3)
Q39	"We hypothesize questions seeking factual information instead of opinions make the dataset less prone to bias." (p.3)
Q74	"In order to test how well models trained on GermanQuAD generalize to other data, we evaluate them on the German part of MLQA (Lewis et al., 2020a), XQuAD (Artetxe et al., 2020), and a private dataset on financial compliance reports called SFCR." (p.6)
Q75	"SFCR is annotated SQuAD-style and consists of 2386 QA pairs." (p.6)
WEB-13	EU Consumer Rights Directive 2011/83/EU (14-day Widerrufsrecht) applies in DE/AT (https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32011L0083)
WEB-14	§ 356a BGB requires a digital Widerrufsbutton with two-step confirmation for German online B2C (https://www.freshfields.com/en/our-thinking/blogs/risk-and-compliance/pitfalls-for-e-commerce-how-the-new-eu-withdrawal-button-widerrufsbutton-wi-102ms91)
WEB-16	Switzerland is not covered by Directive 2011/83/EU; return rights follow Swiss OR and retailer policy (https://www.iubenda.com/en/blog/understanding-the-right-of-withdrawal-in-the-eu-a-guide-for-online-businesses/)

Information Gaps

- No characterization of retailer FAQ/ToS length, sentence structure, or list-density to quantify how far it sits from Wikipedia prose

Requires Expert Verification

- Which specific legal-content categories in the retailer's ToS diverge across DE/AT/CH jurisdictions and would require jurisdiction-aware retrieval

Remediation

Gap 1: Wikipedia-only passages with no retail FAQ/ToS content and no coverage of DE/AT/CH legal variation (§ 356a BGB Widerrufsbutton, Swiss OR return regime)

Recommendation: Assemble a private evaluation corpus from the retailer's own FAQ and ToS, with jurisdiction tags for DE/AT/CH so answers can be evaluated for legal correctness under EU Directive 2011/83/EU [WEB-13, WEB-14] and Swiss OR [WEB-16]. Verify ToS content is current with respect to § 356a BGB.

ID	Evidence
WEB-13	EU Consumer Rights Directive 2011/83/EU (14-day Widerrufsrecht) applies in DE/AT (https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32011L0083)
WEB-14	§ 356a BGB requires a digital Widerrufsbutton with two-step confirmation for German online B2C (https://www.freshfields.com/en/our-thinking/blogs/risk-and-compliance/pitfalls-for-e-commerce-how-the-new-eu-withdrawal-button-widerrufsbutton-wi-102ms91)
WEB-16	Switzerland is not covered by Directive 2011/83/EU; return rights follow Swiss OR and retailer policy (https://www.iubenda.com/en/blog/understanding-the-right-of-withdrawal-in-the-eu-a-guide-for-online-businesses/)

Input Form (IF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: GERMANQUAD_GERMANDPR | Context: German-Speaking E-Commerce Shoppers (DACH)

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark is text-only Standard German in SQuAD JSON format [Q26–Q30], which matches the chatbot's text interface at the modality level. However, annotators were explicitly instructed to reformulate questions with synonyms and altered syntax to reduce lexical overlap [Q21, Q35, Q89] and to write 'very precise natural questions' [Q90] — producing well-formed, formal Standard German. The deployment population types on mobile keyboards with terse, abbreviation-heavy, typo-prone SMS-register German including Umlaut substitutions (ae/oe/ue/ss) and dialect-influenced vocabulary from Austria and Swiss Standard German (register_note in region YAML). Swiss German additionally lacks standardised orthography in informal registers [WEB-1, WEB-2]. Passages are formatted with title prepending [Q28] and length caps [Q29, Q78], which is procedurally similar to what a retrieval pipeline would do — a partial strength. Overall the input-form register gap is substantial.

ID	Evidence
Q21	"To prevent a strong lexical overlap of the question and the corresponding passage, a shortcoming of SQuAD (Kwiatkowski et al., 2019), the annotators reformulate the question text with the use of synonyms and altered syntax wherever possible." (p.2)
Q26	"The basis of GermanQuAD is the German counterpart of English Wikipedia articles' used in SQuAD." (p.2)
Q27	"From the German Wikipedia, we extract each article's text with gensim's WikiExtractor, split it into passages (paragraphs) and exclude passages that are shorter than 500 characters or mainly contain enumerations." (p.2)
Q28	"We add the title of the German Wikipedia page to each passage." (p.2)
Q29	"Since we annotate only one answer per question in the training set and want to reduce the risk of false-negative answers, we limit the length of training passages to 2500 characters." (p.2)
Q30	"For the test set, we keep long passages and only dismiss short ones." (p.2)
Q35	"We continuously check for lexical overlap between question and passage and correct the annotation style in individual sessions with the corresponding annotator." (p.3)
Q78	"For the training, queries are clipped after 32 tokens and passages after 300 tokens. The learning rate is 10–6 using Adam, linear scheduling with warm-up, and dropout rate 0.1. With a batch size of 40, the model achieves an average in-batch rank of 0.50 after training for 20 epochs and the average rank does not further improve after more epochs." (p.6)
Q89	"Do not copy paste answer text into the question. Good questions do not contain the exact same words as the answer or the context around the answer. The question should be a reformulation with synonyms and in different order as the context of the answer." (p.8)
Q90	"Questions should be very precise natural questions you would ask when you want information from another person." (p.8)
WEB-1	Swiss German lacks standardised orthography, creating difficulties for automated text processing (https://www.cl.uzh.ch/en/research-groups/texttechnologies/research/associated/nlp4swissgerman.html)
WEB-2	Modular adaptation work confirms Swiss German written dialect is a distinct NLP challenge (https://arxiv.org/abs/2401.14400)

Strengths

- Text-only modality matches the deployment's chat interface [Q26–Q30]
- Preprocessing choices (title prepending, paragraph splitting, length caps) [Q27–Q30, Q78] are compatible with production retrieval pipelines

- Authors explicitly flag that multilingual models are unsuited for production latency [Q9], which is aligned with deployment constraints

ID	Evidence
Q9	"Multilingual models are much larger than their monolingual counterparts, rendering them unsuitable for most production systems where memory consumption and query speed matter." (p.1)
Q26	"The basis of GermanQuAD is the German counterpart of English Wikipedia articles' used in SQuAD." (p.2)
Q27	"From the German Wikipedia, we extract each article's text with gensim's WikiExtractor, split it into passages (paragraphs) and exclude passages that are shorter than 500 characters or mainly contain enumerations." (p.2)
Q28	"We add the title of the German Wikipedia page to each passage." (p.2)
Q29	"Since we annotate only one answer per question in the training set and want to reduce the risk of false-negative answers, we limit the length of training passages to 2500 characters." (p.2)
Q30	"For the test set, we keep long passages and only dismiss short ones." (p.2)
Q78	"For the training, queries are clipped after 32 tokens and passages after 300 tokens. The learning rate is 10 ⁻⁶ using Adam, linear scheduling with warm-up, and dropout rate 0.1. With a batch size of 40, the model achieves an average in-batch rank of 0.50 after training for 20 epochs and the average rank does not further improve after more epochs." (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal-distribution mismatch: benchmark questions are deliberately reformulated formal Standard German [Q21, Q89, Q90]; deployment inputs are terse, mobile-typed, typo-prone, abbreviation-heavy, and register-mixed queries (region YAML register_note). No Austrian or Swiss variant text is included; no dialect-influenced typed text is included [WEB-1, WEB-2].

IF-2: Regional infrastructure supports the same text-in modality; internet penetration is near-universal across DACH [WEB-4, WEB-5, WEB-6]. Mobile-first access [WEB-6, region YAML dominant_access_mode] introduces typing-behaviour differences the benchmark does not test.

IF-3: Domain-specific form differences: (a) informal SMS-register vs. formal reformulated register; (b) Umlaut substitution (ae/oe/ue/ss) on mobile keyboards; (c) Austrian and Swiss vocabulary variants; (d) missing punctuation and diacritics. None are represented in the benchmark.

IF-4: External-validity harm: model behaviour on formal Wikipedia-style questions may not predict behaviour on terse mobile support queries. This is a documented register gap between test-time and deployment-time input distributions.

ID	Evidence
Q9	"Multilingual models are much larger than their monolingual counterparts, rendering them unsuitable for most production systems where memory consumption and query speed matter." (p.1)
Q21	"To prevent a strong lexical overlap of the question and the corresponding passage, a shortcoming of SQuAD (Kwiatkowski et al., 2019), the annotators reformulate the question text with the use of synonyms and altered syntax wherever possible." (p.2)
Q35	"We continuously check for lexical overlap between question and passage and correct the annotation style in individual sessions with the corresponding annotator." (p.3)
Q89	"Do not copy paste answer text into the question. Good questions do not contain the exact same words as the answer or the context around the answer. The question should be a reformulation with synonyms and in different order as the context of the answer." (p.8)
Q90	"Questions should be very precise natural questions you would ask when you want information from another person." (p.8)

WEB-1	Swiss German lacks standardised orthography, creating difficulties for automated text processing (https://www.cl.uzh.ch/en/research-groups/texttechnologies/research/associated/nlp4swissgerman.html)
WEB-2	Modular adaptation work confirms Swiss German written dialect is a distinct NLP challenge (https://arxiv.org/abs/2401.14400)
WEB-4	Germany 93.5% internet penetration (2024) (https://fred.stlouisfed.org/series/ITNETUSERP2DEU)
WEB-5	Austria 94.9% internet penetration (2024) (https://fred.stlouisfed.org/series/ITNETUSERP2AUT)
WEB-6	Switzerland ~99% internet penetration (Oct 2025) (https://statista.com/statistics/1391386/countries-europe-with-the-highest-internet-penetration-rate)

Information Gaps

- No characterization of typo rate, average query length, or Umlaut-substitution frequency in the deployment population

Requires Expert Verification

- The magnitude of accuracy degradation from formal-to-informal register shift on this specific model, which cannot be estimated from the benchmark alone

Remediation

Gap 1: Formal Standard German reformulated questions do not represent mobile SMS-register, typo-prone, abbreviation-heavy, Umlaut-substituted, or Austrian/Swiss-variant user input

Recommendation: Construct a register-perturbation test set by paraphrasing gold questions into mobile-typed style (short, typo-injected, abbreviated, Umlaut-substituted) and adding Austrian and Swiss vocabulary variants elicited from regional stakeholders; measure accuracy degradation vs. clean queries.

Output Ontology (OO)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: GERMANQUAD_GERMANDPR | Context: German-Speaking E-Commerce Shoppers (DACH)

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output space is extractive text spans — contiguous substrings of the passage — with a distinction between short and long answers [Q23, Q94–Q96]. Questions are required to be self-sufficient [Q24, Q25]. Extractive spans are a plausible primitive for FAQ answer extraction, so the mechanism is partially aligned. However, the ontology has no 'no-answer' / abstain category (SQuAD v2.0 would supply this [WEB-10]) and no escalation-routing category. In a no-human-review deployment where wrong answers are shown directly to customers and misinformation carries GDPR Art. 22 [WEB-20, WEB-21] and EU AI Act exposure [WEB-22], the absence of an abstain output class is a structural-validity gap. Additionally, extractive spans may be inadequate for conditional retail answers that combine multiple ToS clauses (e.g., 'if... then... unless...').

ID	Evidence
Q23	"Consequently, next to answers that are short (a few words, a single number, or an entity), there are also long answers spanning multiple sentences that cannot be shortened without becoming incomplete or even incorrect." (p.2)
Q24	"The questions are self-sufficient in the sense that no additional information is needed to give an answer." (p.2)
Q25	"To ensure self-sufficient questions, we teach the annotators to classify SQuAD questions into self-sufficient ("When did Israel join the OECD?") versus incomplete ("What year did Albert die?")." (p.2)
Q94	"Always mark whole words. Do not start or end the answer within a word." (p.8)
Q95	"For short answers: The answer should be as short and as close to a spoken human answer as possible. Do not include punctuation." (p.8)
Q96	"For long answers: Please mark whole sentences with punctuation. The sentences can also pick up parts of the question, or mark even whole text passages. Mark passages only if they are not too large (e.g. not more than 8-10 sentences)." (p.8)
WEB-10	SQuAD 2.0 defines the unanswerable-question output class the benchmark lacks (https://arxiv.org/abs/1806.03822)
WEB-20	GDPR Art. 22 restricts solely-automated decisions with significant effects (https://gdprinfo.eu/gdpr-article-22-explained-automated-decision-making-profiling-and-your-rights)
WEB-21	E-commerce chatbot GDPR analysis identifies automated FAQ-answering as a candidate Art. 22 concern (https://www.usefini.com/guides/ecommerce-support-bots-gdpr-compliant)
WEB-22	Hamburg DPA 2025 guidance notes AI Act obligations are additive to GDPR Art. 22 (https://kukie.io/blog/automated-decision-making-profiling-gdpr-article-22)

Strengths

- Extractive-span outputs are a reasonable primitive for FAQ answer extraction where answers are literal substrings of source documents [Q23]
- Distinction between short and long answers with explicit annotation guidance [Q94–Q96] accommodates variable answer granularity
- Self-sufficiency requirement [Q24, Q25] aligns with the deployment need for standalone answers shown to customers without context

ID	Evidence
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Q23	"Consequently, next to answers that are short (a few words, a single number, or an entity), there are also long answers spanning multiple sentences that cannot be shortened without becoming incomplete or even incorrect." (p.2)
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Q96	"For long answers: Please mark whole sentences with punctuation. The sentences can also pick up parts of the question, or mark even whole text passages. Mark passages only if they are not too large (e.g. not more than 8-10 sentences)." (p.8)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Categorical review: only one output class — extractive text span [\[Q23\]](#). No abstain, no escalate, no 'requires human review' class.

OO-2: Missing categories for the DACH deployment: (a) abstain/no-answer when the FAQ/ToS does not contain the information; (b) escalation to live agent; (c) potentially a 'jurisdiction-specific' flag for DE/AT/CH-varying content [\[WEB-16\]](#).

OO-3: No categories encode explicitly non-regional values, but the assumption that every well-formed question has a supported answer span [\[Q24\]](#) embeds a values-choice (favouring answer-generation over abstention) that is misaligned with the deployment's safety needs.

OO-4: Stakeholder-driven redesign is warranted: adding an unanswerable class and an escalation class, following the SQuAD v2.0 pattern [\[WEB-10\]](#) and the MÖVE FAQ-oriented approach [\[WEB-12\]](#).

OO-5: Structural-validity harm: the benchmark's output space cannot represent operationally critical decisions (abstain, escalate), so passing the benchmark provides no evidence about these behaviours.

ID	Evidence
Q23	"Consequently, next to answers that are short (a few words, a single number, or an entity), there are also long answers spanning multiple sentences that cannot be shortened without becoming incomplete or even incorrect." (p.2)
Q24	"The questions are self-sufficient in the sense that no additional information is needed to give an answer." (p.2)
Q25	"To ensure self-sufficient questions, we teach the annotators to classify SQuAD questions into self-sufficient ("When did Israel join the OECD?") versus incomplete ("What year did Albert die?")." (p.2)
WEB-10	SQuAD 2.0 defines the unanswerable-question output class the benchmark lacks (https://arxiv.org/abs/1806.03822)
WEB-12	MÖVE uses embedding- and LLM-based metrics for German FAQ-style QA — an alternative metric approach relevant to the deployment (https://arxiv.org/abs/2606.13111)
WEB-16	Switzerland is not covered by Directive 2011/83/EU; return rights follow Swiss OR and retailer policy (https://www.iubenda.com/en/blog/understanding-the-right-of-withdrawal-in-the-eu-a-guide-for-online-businesses/)
WEB-20	GDPR Art. 22 restricts solely-automated decisions with significant effects (https://gdprinfo.eu/gdpr-article-22-explained-automated-decision-making-profiling-and-your-rights)

WEB-22	Hamburg DPA 2025 guidance notes AI Act obligations are additive to GDPR Art. 22 (https://kukie.io/blog/automated-decision-making-profiling-gdpr-article-22)
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Information Gaps

- Baseline rate of unanswerable queries expected in the deployment (region YAML does not quantify this)

Requires Expert Verification

- Whether the retailer's chatbot architecture wraps the extractive model with a downstream abstain/escalate classifier, which would partially mitigate this ontology gap

Remediation

Gap 1: Output space contains only extractive spans; no abstain or escalation-routing class in a no-human-review deployment with GDPR Art. 22 and EU AI Act exposure

Recommendation: Introduce an abstain output class and an escalation output class, and evaluate the pipeline on gold-labelled no-answer and escalation cases. Align system design with the GDPR Art. 22 'meaningful human involvement' standard [WEB-20, WEB-21, WEB-22] by ensuring the abstain class triggers human review.

ID	Evidence
WEB-20	GDPR Art. 22 restricts solely-automated decisions with significant effects (https://gdprinfo.eu/gdpr-article-22-explained-automated-decision-making-profiling-and-your-rights)
WEB-21	E-commerce chatbot GDPR analysis identifies automated FAQ-answering as a candidate Art. 22 concern (https://www.usefini.com/guides/ecommerce-support-bots-gdpr-compliant)
WEB-22	Hamburg DPA 2025 guidance notes AI Act obligations are additive to GDPR Art. 22 (https://kukie.io/blog/automated-decision-making-profiling-gdpr-article-22)

Output Content (OC)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: GERMANQUAD_GERMANDPR | Context: German-Speaking E-Commerce Shoppers (DACH)

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark's answer labels are produced through a moderately rigorous multi-annotator process: three-way test annotation with students, an expert, and Crowd Guru workers [Q41–Q47], with detailed guidance and feedback loops [Q17, Q32–Q34]. Annotators include men and women and native and non-native German speakers [Q38], with explicit gender-inclusive-language guidance [Q40]. However, annotators are computational-linguistics/CS/chemistry students plus one NLP expert [Q18] — not shoppers, not CS agents, and not identified as coming from Austria or Switzerland. Their judgments about what constitutes the 'best' answer span for a Wikipedia paragraph are unlikely to correlate strongly with what a DACH online shopper would judge to be the useful answer to a return-eligibility question. Training data is one-way annotated, which the authors acknowledge limits metric reliability [Q53, Q76]. Convergent validity for the deployment's regional user judgments is therefore weakly supported.

ID	Evidence
Q17	"We select annotators that are familiar with machine learning concepts, teach them the use of BERT for QA, give detailed labelling instructions, continuously monitor the annotations, and give feedback wherever necessary." (p.2)
Q18	"We collaborate with students in the area of computational linguistics, computer science, and chemistry and one expert annotator who created more than 20k QA annotations." (p.2)
Q32	"An expert annotator manually checks the first 300 questions and answers for each annotator and gives detailed feedback." (p.3)
Q33	"In weekly group meetings, a random subset of newly created labels is discussed." (p.3)
Q34	"We encourage working in groups to debate imprecision in question formulation or answer selection." (p.3)
Q38	"The team of annotators includes men and women, and native as well as non-native German speakers." (p.3)
Q40	"Since the gender form is a specific issue in the German language, we encourage to include male and female or gender-neutral forms when formulating questions." (p.3)
Q41	"We construct a three-way annotated test set including 2204 questions." (p.3)
Q42	"Student annotators formulate all questions and mark the first answer." (p.3)
Q43	"The second answer comes from our expert annotator who also removes questions that are not formulated clearly." (p.3)
Q44	"For the third annotation, we use the crowd annotation provider Crowd Guru." (p.3)
Q45	"Crowd Guru's annotations are checked for positional overlap with existing answers." (p.3)
Q46	"If there is no overlap, we manually check for irregularities and as a result, remove 76 wrong answers." (p.3)
Q47	"The final dataset consists of 2204 questions with $2204 \cdot 3 - 76 = 6536$ answers." (p.3)
Q53	"As the training dataset is one-way annotated, there is exactly one answer per question, resulting in 11518 question/answer pairs." (p.3)
Q76	"Traditional QA metrics, such as Exact Match and F1, are unsuited for datasets that do not have multi-way annotations because they cannot cover all possible correct answers." (p.6)

Strengths

- Three-way test-set annotation is a meaningful methodological improvement over single-annotator QA benchmarks [Q41–Q47, Q54]

- Explicit quality-control processes: first-300 expert check per annotator, weekly group meetings, positional-overlap validation of crowd labels [Q32–Q34, Q45–Q46]
- Annotator diversity attended to for gender and native/non-native status [Q38, Q40]
- Authors are transparent about training-set single-annotation limitation and its implications for EM/F1 reliability [Q53, Q76]

ID	Evidence
Q32	“An expert annotator manually checks the first 300 questions and answers for each annotator and gives detailed feedback.” (p.3)
Q33	“In weekly group meetings, a random subset of newly created labels is discussed.” (p.3)
Q34	“We encourage working in groups to debate imprecision in question formulation or answer selection.” (p.3)
Q38	“The team of annotators includes men and women, and native as well as non-native German speakers.” (p.3)
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Q53	“As the training dataset is one-way annotated, there is exactly one answer per question, resulting in 11518 question/answer pairs.” (p.3)
Q54	“In contrast to that, the test dataset contains three answers per question: For 2204 questions, there are 6536 answers.” (p.3)
Q76	“Traditional QA metrics, such as Exact Match and F1, are unsuited for datasets that do not have multi-way annotations because they cannot cover all possible correct answers.” (p.6)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: Ground-truth labels reflect NLP-student/expert judgments about Wikipedia answer spans, not DACH shopper or CS-agent judgments. Convergent validity for the deployment population is weak.

OC-2: Disagreement likely: shoppers may prefer terser, more actionable spans; CS agents may prefer spans that include conditional qualifiers. Neither preference profile is represented in the annotator pool [Q18].

OC-3: Annotator demographics documented: field of study, gender diversity, native/non-native German speakers [Q18, Q38]. Country of residence (DE vs. AT vs. CH), age, and shopper/CS-agent status are not documented — a datasheet-level gap.

OC-4: Re-annotation by shoppers or CS agents on retail-FAQ content is warranted for deployment; the current pool cannot be assumed representative.

OC-5: Aggregation for the test set retains all three annotations [Q47, Q54] rather than collapsing to majority, which preserves annotator variance — a positive feature. Crowd Guru answers without positional overlap were removed after manual review, with 76 removed [Q45, Q46]; this may erase legitimate minority interpretations.

OC-6: Documented label-quality issues: (1) demographic gap between annotators and target users; (2) potential erasure of minority interpretations via the 76-answer removal step [Q46]; (3) training set is single-annotated [Q53].

ID	Evidence
Q18	"We collaborate with students in the area of computational linguistics, computer science, and chemistry and one expert annotator who created more than 20k QA annotations." (p.2)
Q38	"The team of annotators includes men and women, and native as well as non-native German speakers." (p.3)
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Q76	"Traditional QA metrics, such as Exact Match and F1, are unsuited for datasets that do not have multi-way annotations because they cannot cover all possible correct answers." (p.6)

Information Gaps

- Annotator country of residence (DE/AT/CH split) is not documented
- No inter-annotator agreement statistics reported at the item level

Requires Expert Verification

- Whether shopper vs. NLP-student answer-span preferences actually diverge on procedural retail content — requires a small annotation study on the retailer's own FAQ

Remediation

Gap 1: Annotator pool is NLP-adjacent students plus one expert, not shoppers or CS agents; country distribution (DE/AT/CH) not documented

Recommendation: Recruit a small representative annotator panel (mix of shoppers and CS agents across DE, AT, and CH) to re-annotate a sample of retailer FAQ/ToS QA pairs; measure agreement with the current gold to quantify convergent-validity risk.

Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: GERMANQUAD_GERMANDPR | Context: German-Speaking E-Commerce Shoppers (DACH)

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark reports Exact Match, F1, and Top-N-Accuracy [Q56, Q66–Q69], with Top-1-Accuracy [Q62] being the most deployment-relevant since the retailer shows one answer to the customer without human review. A human baseline is computed on the three-way test set [Q73]. The authors themselves flag that EM/F1 penalise semantically-correct-but-verbally-extended answers and call for semantic-similarity-aware metrics [Q76, Q83–Q85]. For the deployment, the benchmark’s output form (text span with token-overlap scoring) partially matches (text output is correct modality), but the metric set does not measure abstain accuracy, escalation appropriateness, or customer task-completion — all operationally critical. Top-N-Accuracy’s binary-overlap definition [Q67, Q68] is more forgiving of annotation-style variance and thus a partial mitigation for the register-mismatch risk, but not a substitute for outcome metrics.

ID	Evidence
Q56	“Further, it shows the average answer length (and standard deviation) and the model performance with regard to exact match, F1-score, and Top-1-accuracy.” (p.3)
Q62	“Figure 3 shows the impact of the number of training samples on the percentage of exact matches, F1-score, and Top-1-accuracy (any overlap of prediction and ground truth) on the GermanQuAD test set. For all three metrics, there are no improvements after using more than 80 percent of the training samples.” (p.4)
Q66	“Next to exact match (EM) and the ratio of overlapping words (F1) metric we also report Top-N-Accuracy scores.” (p.5)
Q67	“Top-N-Accuracy is a binary hit or miss score for the first N model predictions.” (p.5)
Q68	“The value is one if there is any positional overlap between the ground-truth answer and the model prediction — otherwise, it is zero.” (p.5)
Q69	“Such a hit or miss score is less prone to details of how answers are annotated (e.g., Answer1: “19th century”, Answer2: “middle of 19th century”) and useful in settings without multi-way annotations or where answers can be more than numbers, single entities, or a few words.” (p.5)
Q73	“Table 4: QA model performance on the three-way annotated GermanQuAD test set (in percent). Model types and training data are included in the model name. For finetuning XLM-Roberta we use the English SQuAD v2.0 dataset, the GELECTRA models are warm started on the German translation of SQuAD v1.1 and finetuned on GermanQuAD. The human baseline was computed for the 3-way test set by taking one answer as prediction and the other two as ground truth.” (p.6)
Q76	“Traditional QA metrics, such as Exact Match and F1, are unsuited for datasets that do not have multi-way annotations because they cannot cover all possible correct answers.” (p.6)
Q83	“A promising path for future work is to research new evaluation metrics for QA.” (p.7)
Q84	“Especially for longer answers in German, we find that current metrics penalise too much if a predicted answer contains some extra words although being semantically similar to the ground-truth answer.” (p.7)
Q85	“Future metrics could take into account the semantic similarity of predicted and ground-truth answers.” (p.7)

Strengths

- Top-1-Accuracy is reported and is the deployment-relevant metric for no-human-review chat output [Q62]
- Human baseline on the three-way test set provides a comparison target [Q73]
- Top-N-Accuracy’s overlap-based definition [Q67–Q69] partially mitigates EM/F1 sensitivity to annotation-style variation for long answers

- Authors are transparent about EM/F1 limitations and identify semantic-similarity metrics as future work [Q76, Q83–Q85]

ID	Evidence
Q62	“Figure 3 shows the impact of the number of training samples on the percentage of exact matches, F1-score, and Top-1-accuracy (any overlap of prediction and ground truth) on the GermanQuAD test set. For all three metrics, there are no improvements after using more than 80 percent of the training samples.” (p.4)
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Q85	“Future metrics could take into account the semantic similarity of predicted and ground-truth answers.” (p.7)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (text) matches the chat interface. Metric coverage is incomplete: EM/F1 penalise semantically-correct answers [Q76, Q84]; abstain accuracy and escalation accuracy are entirely unmeasured; task-completion rate is not measured.

OF-2: Not applicable — the deployment is text-based chat, not voice. No TTS requirement.

OF-3: Literacy is near-universal across DACH (~99% general functional literacy [WEB-3]), though PIAAC indicates ~17–18% of adults have basic-or-below reading proficiency, which may affect comprehension of formal FAQ/ToS answers extracted verbatim. The benchmark does not evaluate answer readability or plain-language suitability.

OF-4: External-validity harm: passing EM/F1/Top-N on Wikipedia QA does not entail deployment-time correctness (semantic-similarity gap [Q84]) or safety (no abstain metric). The MÖVE benchmark [WEB-12] uses embedding- and LLM-based metrics more suitable for procedural FAQ content, which is a documented alternative for this deployment.

ID	Evidence
Q62	“Figure 3 shows the impact of the number of training samples on the percentage of exact matches, F1-score, and Top-1-accuracy (any overlap of prediction and ground truth) on the GermanQuAD test set. For all three metrics, there are no improvements after using more than 80 percent of the training samples.” (p.4)
Q66	“Next to exact match (EM) and the ratio of overlapping words (F1) metric we also report Top-N-Accuracy scores.” (p.5)

Q76	"Traditional QA metrics, such as Exact Match and F1, are unsuited for datasets that do not have multi-way annotations because they cannot cover all possible correct answers." (p.6)
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Q84	"Especially for longer answers in German, we find that current metrics penalise too much if a predicted answer contains some extra words although being semantically similar to the ground-truth answer." (p.7)
Q85	"Future metrics could take into account the semantic similarity of predicted and ground-truth answers." (p.7)
WEB-3	OECD PIAAC indicates ~17–18% of DACH adults have basic-or-below reading proficiency (https://www.oecd.org/skills/piaac/)
WEB-12	MÖVE uses embedding- and LLM-based metrics for German FAQ-style QA — an alternative metric approach relevant to the deployment (https://arxiv.org/abs/2606.13111)

Information Gaps

- No task-completion, customer-satisfaction, or escalation-appropriateness metric is defined by the benchmark, and no proxy is documented
- Answer-readability / plain-language suitability of extracted spans is not evaluated

Requires Expert Verification

- Whether the retailer's chatbot pipeline can wrap the extractor with a downstream semantic-similarity or LLM-judge metric (as MÖVE demonstrates [WEB-12]) to close the metric gap

Remediation

Gap 1: EM/F1 penalise semantically-correct extended answers [Q84]; no metric measures abstain accuracy, escalation accuracy, or task completion

Recommendation: Adopt MÖVE-style embedding- and LLM-based semantic-similarity metrics [WEB-12] alongside Top-1-Accuracy, and add abstain-precision/recall and escalation-appropriateness metrics on the supplementary evaluation set. Track deployment-time task-completion via product telemetry as an external validation signal.

ID	Evidence
Q84	"Especially for longer answers in German, we find that current metrics penalise too much if a predicted answer contains some extra words although being semantically similar to the ground-truth answer." (p.7)
WEB-12	MÖVE uses embedding- and LLM-based metrics for German FAQ-style QA — an alternative metric approach relevant to the deployment (https://arxiv.org/abs/2606.13111)